ARASTU THAKUR

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AI/ML ENGINEER | GENERATIVE AI | SOFTWARE ENGINEERING | AUTOMATION

PROFESSIONAL SUMMARY

AI/ML Engineer with 2+ years of industry experience at Intel building production machine learning systems, Generative AI applications, automation solutions, and data-driven engineering workflows. Strong experience in Python, C++, SQL, PyTorch, TensorFlow, LLMs, LangChain, LangGraph, Docker, Kubernetes, CI/CD, and MLflow. Experienced in developing ML pipelines, LLM-powered retrieval systems, deep learning solutions, validation automation, and scalable analytics systems. Proven track record of reducing validation effort, improving engineering efficiency, and delivering production-focused AI and software solutions.

SKILLS

Programming: Python, C++, SQL, Bash, MATLAB
AI & Machine Learning: Deep Learning, PyTorch, TensorFlow, Transformers, Autoencoders, Self-Attention, Computer Vision
Generative AI: LangChain, Retrieval-Augmented Generation (RAG), Embeddings, Vector Databases, Agentic AI
Data & Analytics: SQL, Pandas, NumPy, Data Analysis, Big Data Analytics, Data Pipelines
MLOps & DevOps: MLflow, Docker, Kubernetes, CI/CD, Git, GitLab, GitLab CLI, CMake, Linux
Software Engineering: Object-Oriented Programming, REST APIs, Software Design, Modular Development, Automation

PROFESSIONAL EXPERIENCE

Data Scientist — Intel, Bengaluru, India
Jun 2025 – Present

- Developed AI-driven automation and deep learning-based encoder solutions for defect reduction, achieving ~94% validation accuracy and improving testing efficiency by 10%.
- Built and maintained production ML automation and SQL-based big data analytics systems supporting 20+ users and processing ~1,000+ queries daily across multiple engineering workflows.

Graduate Technical Intern — Intel, Bengaluru, India
Jul 2024 – Jun 2025

- Developed AI-powered automation bots and debugging utilities that reduced repetitive validation efforts by ~30% and improved diagnostics accuracy across engineering workflows.
- Migrated and optimized 1,200+ validation test cases while supporting SQL-driven big data analytics and reporting pipelines for large-scale validation operations.

PROJECTS

Ever Ready to Release (Intel | Internal)
Tech: Python, LangChain, LLMs, Vector Databases, C++, MLflow

- Architected and deployed an LLM-powered retrieval workflow using LangChain to map code changes with relevant validation test cases, reducing validation time by 30% and infrastructure usage by ~20%.
- Built scalable AI-assisted automation pipelines integrating embeddings, semantic search, and automated validation workflows.

AI-Powered Data Analysis & Insights Platform (GitHub)
Tech: Python, Streamlit, TensorFlow, Scikit-learn, XGBoost, LightGBM, Gemini AI, Plotly

- Developed an end-to-end AI-powered analytics platform integrating machine learning, deep learning, and Gemini-powered insight generation for automated exploratory data analysis.
- Engineered scalable ML workflows supporting automated preprocessing, visualization, model training, hyperparameter optimization, and performance evaluation across structured datasets.

MRI Image Denoiser (GitHub)
Tech: TensorFlow, Autoencoders, Self-Attention

- Designed a residual autoencoder with self-attention mechanisms for MRI image denoising, achieving ~18% PSNR improvement.
- Optimized inference performance to under 200 ms per image, enabling efficient deployment in medical environments.

EDUCATION

Bachelor of Technology in Computer Science — Jain (Deemed-to-be) University | Sep 2021 – May 2025
Cumulative GPA: 9.757/10 | Gold Medalist

AWARDS & CERTIFICATES

- Secured full-tuition Study in India Scholarship (Top 5% of 10,000+ applicants)
- 1st Runner-Up – JAIN x Honeywell Hackathon 2024 (Fire-Safe Environment theme)
- Advanced Machine Learning Operations – Duke University
- Generative AI Engineering with LLM – IBM

EXTRA-CURRICULAR ACTIVITIES

- Microsoft Student Ambassador with experience mentoring students in AI and software engineering initiatives.
- Over 3 years of research experience resulting in 15 publications with 550+ citations, with the highest impact factor of **10.9**.